

TRIAD: Trusted RDF Interoperable Agent Data

Asish A. George
University of Victoria
Canada
asishgeorge@uvic.ca

Mohamed Khaled Abuela
University of Victoria
Canada
mabuela@uvic.ca

Abstract

Patients can wait and deteriorate while information about them is missed, delayed, or scattered across records, and physicians spend much of their time piecing it together. We propose TRIAD, a data model that connects the information recorded by nurses and physicians so it can be shared and updated as it changes, and that allows an agent to retrieve that information directly rather than search for it. Code: github.com/MohamedKhalidmk/TRIAD (CC BY-NC-SA 4.0).

Keywords

Knowledge Graph, FHIR, Agent Memory, Privacy

1 Introduction

When a patient calls a clinic, the information they give may be incomplete or inaccurate, and it is then summarized by a nurse. Patients may be prioritized incorrectly, so urgent cases wait; reviews of emergency triage report nurse accuracy between 59.3% and 82% [4]. Physicians also lose time on wrong or outdated information and spend 49% of their office day on electronic records and desk work [3].

Information about a patient comes from several sources over time: the patient's report, the provincial pharmacy record (PharmaNet), test results, the nurse's assessment and recommendation (SBAR), and the physician's decision to accept, reject, or build on it. Because different people and organizations create this history, records can disagree. The physician then cannot easily tell whether a record reflects what the patient is actually taking, who recorded and verified it, and whether the nurse's recommendation was accepted.

Existing standards such as HL7 FHIR [1] can record where each piece of information came from, but they point at conflicts rather than resolve them. A patient's medication information is scattered across orders, dispenses, administrations, patient reports, and claims, and clinicians must reconcile it by hand, which takes time away from direct patient care [2].

2 Motivation

Hospitals are slow to use agents in clinical work, for two reasons. First, an agent cannot keep all of this data in its conversation window. In a long context, the agent can miss that a fact changed and use the old fact.

Second, patient data is sensitive. Personally Identifiable Information (PII), such as a name or a health card number, becomes Protected Health Information (PHI) when it is linked to health data. Laws such as HIPAA in the United States, and PIPEDA and PHIPA in Canada, control the use of PHI. A hospital cannot use an agent that shows PHI to the wrong person or does not obey these laws.

An Electronic Health Record (EHR) is designed for people to read. It does not show the agent who asserted what and what replaced what, so the agent must infer this at runtime. When the inference is wrong, nobody can find the cause, staff stop trusting the agent, and hospitals do not use it.

2.1 Motivating Example

A patient comes to the emergency room with shortness of breath. The nurse assesses the patient and records a possible asthma attack. One hour later, the physician examines the patient and revises the assessment to possible heart failure. The physician also tells the agent that the father of the patient died of a heart attack. This fact is not in the EHR yet.

In the evening, the patient asks the agent about the plan for their care. The conversation is now long. The asthma assessment appears many times, and the revision appears once. The agent answers from the old assessment and tells the patient that an inhaler is the plan.

With TRIAD, the agent starts at the patient and finds the current assessment one or two hops away in an RDF [5] knowledge graph. It sees that the physician replaced the assessment of the nurse, and it finds the family history. If its answer is wrong, we can trace the error to a specific relationship. The agent shows the patient only approved data, and staff who do not care for this patient get no access to their PHI.

3 Team Justification

The project combines Mohamed Khaled Abuela's M.Sc. thesis research on memory for language model agents, under Professor Shengyao Lu, with Asish George's four years of experience developing healthcare systems, so the problem comes from real work. Abuela wrote the abstract, introduction, and references and set up the repository. George wrote the motivation and example, created the name and logo, and proposed FHIR. Both edited the proposal.

References

- [1] HL7 International. 2023. FHIR: Fast Healthcare Interoperability Resources. <https://hl7.org/fhir/>.
- [2] Institute for Safe Medication Practices Canada. [n. d.]. Medication Reconciliation (MedRec). <https://www.ismp-canada.org/medrec/>.
- [3] Christine Sinsky et al. 2016. Allocation of Physician Time in Ambulatory Practice: A Time and Motion Study in 4 Specialties. *Annals of Internal Medicine* 165, 11 (2016), 753–760.
- [4] Krisada Suamchaiyaphum, Allison R. Jones, and Adelais Markaki. 2024. Triage Accuracy of Emergency Nurses: An Evidence-Based Review. *Journal of Emergency Nursing* 50, 1 (2024), 44–54.
- [5] W3C. 2014. RDF 1.1 Concepts and Abstract Syntax. <https://www.w3.org/TR/rdf11-concepts/>.

A Original Work Before AI-Assisted Editing

Per the Level C guideline, this appendix contains both authors' original text, written without AI, before AI-assisted editing (A.1: Asish A. George, a spoken transcript with speech errors removed; A.2 onward: Mohamed Khaled Abuela). AI (Claude) was also used to locate and explain sources.

A.1 Motivation: Original Transcript

Spoken transcript by Asish A. George. Filler words, repeated words, and speech-to-text errors are removed. The wording is otherwise unchanged.

I want to talk about the motivation for the project. There are different levels of motivation. First, in multi-agent systems, let's say a doctor has an agent to talk about patient data, and a nurse has an agent to talk about the patient data. These two agents may have fragmented memory about the patient data itself. We don't want different people to have fragmented memory about the same element. We want to have a cohesive memory within an agentic system so that patient care happens in a much easier way.

That aside, agentic systems are not being accepted or adopted into real healthcare systems as much, for different reasons. One is that the data is fragmented across different systems. The second is the concern of privacy, whether a data leak will happen. And third, there are strict regulations like HIPAA, and PHIPA in Canada, where we have to adhere to certain laws to support the healthcare environment.

We want to create a system or a data model that solves most of these problems. In an agentic system, we want a really fast lookup. So having a triple store, or RDF-based store, where we can query a patient's data would be beneficial for us.

The second is in terms of the data store. Like I said, there are multiple fragmented data stores, but the most common is FHIR data. It might not be widely adopted, but it is the global standard for interoperability for healthcare data. So we could model the agent in such a way that any FHIR data source can be attached to the agent, so that we can load FHIR data into the RDF store at any time, and the agent can operate within that data store.

The third is in terms of data privacy. Even though we are using a graph database to hold the relations, we also try to ensure that no data leaks, that one patient's data is not shown to another patient, and that only the practitioners who are responsible for the patient can see the patient.

This is the motivation behind the project and what we want to do with it.

A.2 Abstract

patient wait and goes critical and miss information keeps delaying while physicians is working 24/7 and we are just trying to make standards that arent helping but just pointing triad automates the entire processes from summary to giving the doctor what he needs using a smart data model that doesnt require the agent to ask questions and fetch but just to pull without reason sharing across nurse and doctor info can be shared and exchanged and edited right away without any problems all connected

A.3 Problem Definition Notes

Answers to the six problem-definition questions, as originally written:

- (1) patient on phone then pharmacy then xrays then ISMP then SBAR
- (2) patient history and medications different other clinics or out of data, also name different in entities like m.khaled or mohamed khaled or different descriptions of symptoms
- (3) what patient who wrote it who verified
- (4) Next patient, condition, how urgent and what they are currently taking and nurse recommendation
- (5) same
- (6) why it was recorded who recorded it when and patient full history and condition

Follow-up answers: medications the patient currently taking and it gives the doctor an idea and the doctor can use it or reject it or expand it; history and results; human symbols and writing and also what patient and time and author revision verification etc.

Problem paragraph: Patient history can differ as it's created by different people and organisation and it's not easy to know without wasting the doctors time also a nurse can miss important info over the phone and the doctor can waste some time on something he didn't need to do while also other more important patient suffer

A.4 Introduction: Opening

so the patient when he calls a hospital the info he gives might not be that accurate or summaries so the nurse summaries it [...] that has some serious problems which are first the ordering of patients are incorrect, causing important patients to wait critical time and see that the doctor time is wasted with completely wrong info, also doctors deal with the slowest out dated systems that doesn't just give the info right away we need a source for that

A.5 Introduction: Disagreement Between Records

Because this history is created by different people and organizations, records can disagree, and the same patient may appear under different names. When records disagree, the physician cannot easily tell whether a record still reflects what the patient is actually taking, who recorded it and when, who verified it, and whether the nurse's recommendation was accepted, rejected, or built on.

A.6 Introduction: Gap

fhir just signifies what comes from a different source not resolve the conflict it's pointing at the problem not offering solutions and also the scattershot thing I said and also most of clinical work is still manual and you can find many evidence